

1. Administrative & Study Identification

Full Study Title: A Multiple-center, Non-randomized, Open-label, Adaptive, Single-ascending Dose (Part 1 and Part 2) and Multiple-ascending Dose (Part 3) Parallel, Phase IB Study to Investigate the Safety, Tolerability, Immunogenicity, Pharmacokinetics, and Pharmacodynamics of R07121932 Following Intravenous (Parts 1) and Subcutaneous Administration (Parts 2 and 3) in Participants With Multiple Sclerosis **Short Title/Acronym:** B-Shuttle Clinical Trial **Protocol Number:** BP42230 **NCT Number:** NCT05704361 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** R07121932 (Brain Shuttle Anti-CD20 Monoclonal Antibody) **Phase of Development:** Phase I (Phase IB) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of a single-ascending intravenous (IV) dose (Part 1), a single-ascending subcutaneous (SC) dose (Part 2), and multiple ascending SC doses (Part 3) of R07121932 in participants with multiple sclerosis (MS). **Secondary Objectives:** To investigate the immunogenicity, pharmacokinetics (PK), and pharmacodynamics (PD) of R07121932, specifically to observe how the drug gets to different parts of the body, how the body processes and eliminates it, and how it affects the immune system.

2.2 Background & Rationale While disease-modifying therapies (DMTs) are effective at reducing MS activity, they have difficulty penetrating the central nervous system (CNS) due to the blood-brain barrier. R07121932 is an experimental Type 2 anti-CD20 monoclonal antibody that uses a transferrin-based "brain shuttle technology" to pass through the blood-brain barrier. The goal is to evaluate if this enhanced CNS penetration can eradicate CNS-resident B-cells, potentially stopping nerve damage and further reducing confirmed disability progression (CDP) beyond what peripherally restricted therapies can achieve.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Non-randomized) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **N**: 129 participants
Number of Sites: Multiple centers globally (including the USA)
Estimated Study Start: 2021 **Estimated Primary Completion**: Jul 7, 2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 65 years, inclusive. 2. Participants with relapsing multiple sclerosis (RMS) or progressive multiple sclerosis (PMS) (including RRMS, PPMS, and SPMS) who fulfil international panel criteria for diagnosis (McDonald 2017 criteria). 3. Expanded Disability Status Scale (EDSS) score ≤ 7.0 at Screening. 4. Participants not treated with any approved MS treatment at Screening and not planning to start on any MS therapy during the study.

4.2 Exclusion Criteria 1. Evidence of clinical disease activity defined by any clinical relapse within 3 months prior to screening, or >1 clinical relapse within 12 months prior to screening. 2. Evidence of magnetic resonance imaging (MRI) activity as defined by the presence of ≥ 1 Gadolinium (Gd)-enhancing T1 lesion or ≥ 4 new/enlarging T2 lesions. 3. Active, confirmed, or high degree of suspicion for progressive multifocal leukoencephalopathy (PML). 4. Previous treatment with anti-CD20 B-cell-depleting therapies (e.g., rituximab, ocrelizumab, or ofatumumab) <12 months prior to screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single-ascending dose (Part 1 and 2) or multiple-ascending dose (Part 3) administered according to adaptive cohort escalation. **Route of Administration**: Intravenous (IV) infusion (Part 1) and Subcutaneous (SC) injection (Parts 2 and 3). **Duration of Treatment**: Total time of participation is about 7 to 8 months, which includes an approximate 6-month post-treatment follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Physiotherapy or continuation of symptomatic treatments for MS that were already started prior to screening. **Prohibited**: Any approved MS disease-modifying therapies (DMTs) at Screening or during the study; any concomitant disease requiring systemic corticosteroids or immunosuppressants; vaccination with live/live-attenuated vaccines within 6 weeks prior to Day 1.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to Day -1 Informed Consent,

EDSS assessment, Baseline MRI, Evaluation of MS Treatment History, Safety Labs | | **Baseline** | Day 1 | First Dose of R07121932 (IV or SC depending on cohort) | | **Interim** | Over 6 Months | Regular Safety Assessments, PK/PD Blood and optional CSF sampling, Symptom Monitoring | | **End of Study** | Month 6 (Post-Tx)| Final Follow-up Visit, Assessment of Unwanted Effects, Exit Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The number and seriousness of any unwanted effects (safety and tolerability) during treatment and the 6-month follow-up period. **Measurement Method:** Clinical observation, safety laboratory results, and adverse event reporting system per standard trial protocols.

7.2 Secondary & Exploratory Endpoints * **Pharmacokinetics (PK):** Measurement of R07121932 concentration in different parts of the body (blood and CSF) to assess absorption, distribution, metabolism, and excretion. * **Pharmacodynamics (PD):** Assessment of how R07121932 affects the immune system (e.g., B-cell depletion levels in blood and CNS compartments). * **Immunogenicity:** Incidence and levels of anti-drug antibodies against R07121932.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Participants are closely monitored during the trial with regular safety assessments for mild to life-threatening side effects, including infusion/injection-related reactions. **Serious Adverse Events (SAEs):** Standard protocol for immediate reporting of life-threatening events or occurrences requiring hospitalization. **Data Monitoring Committee (DMC):** Managed via an independent data safety monitoring board to evaluate dose-escalation safety thresholds in this adaptive design.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set will include all participants who receive at least one dose of the investigational product. **Sample Size Justification:** \$N=129\$ participants, estimated based on adaptive, dose-escalation phase I requirements for adequate safety, tolerability, and PK/PD profiling across IV and SC cohorts. **Handling of Missing Data:** Standard handling procedures for early-phase, open-label adaptive trials will be employed for missing PK/PD or safety data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency on-site monitoring typical for first-in-human or early phase brain-shuttle therapies, tracking both systemic and CNS-specific safety parameters. **Audit Trail:** Robust tracking via eCRF locking, and rigorous auditing of clinical and MRI endpoints.

11. Study Identifiers & Governance

Primary ID: BP42230 **Registry Number:** NCT05704361 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** A Study to Investigate the Safety, Tolerability, and Processing by the Body of Intravenous and Subcutaneous R07121932 Administration in Participants With Multiple Sclerosis **Official Regulatory Title:** A Multiple-center, Non-randomized, Open-label, Adaptive, Single-ascending Dose (Part 1 and Part 2) and Multiple-ascending Dose (Part 3) Parallel, Phase IB Study to Investigate the Safety, Tolerability, Immunogenicity, Pharmacokinetics, and Pharmacodynamics of R07121932 Following Intravenous (Parts 1) and Subcutaneous Administration (Parts 2 and 3) in Participants With Multiple Sclerosis

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Multiple Sclerosis (Relapsing and Progressive forms; RRMS, SPMS, PPMS) **Diagnosis Threshold:** Fulfilling the McDonald 2017 criteria with an EDSS ≤ 7.0 **Medical**

Methodology: Investigational Disease-Modifying Therapy (Brain Shuttle Anti-CD20 Monoclonal Antibody) **Optimization Target:** Targeting CNS resident B-cells traversing the blood-brain barrier to halt nerve damage and confirmed disability progression

13. Study Procedures & Methodology

Management Pathway: Specialized Phase I unit pathway focusing on ascending dosing and rigorous safety monitoring. **Education**

Protocol (HCP): Extensive protocol training regarding brain-shuttle technology administration and specialized CSF/blood sample handling. **Education Protocol (Patient):** Strict guidelines provided on identifying signs of infusion/injection reactions, relapse symptoms, and managing MS symptoms during the trial.

Quality Audit Mechanism: Continuous safety data reviews prior to advancing to higher dosage cohorts in the adaptive model.

14. Observation & Follow-Up Schedule

Total Duration: 7 to 8 months per participant **Onsite Visit**

Cadence: Frequent visits during the dosing phase (Part 1/2 single dose, Part 3 multiple dose) followed by periodic evaluation.

Remote Monitoring Frequency: Supplemental virtual health checks and diary symptom monitoring where standard clinical practice allows.

Checklist Review Interval: Continuous throughout the 6-month safety follow-up period.

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					